

VINCENT BIO BEDIAKO

Curriculum Vitae

Graduate Group in Demography · University of Pennsylvania

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PROFESSIONAL SUMMARY

Demographer, data scientist, and researcher with 7+ years of combined experience in statistical analysis, data visualization, and research implementation. Proven expertise in developing data collection strategies, implementing advanced statistical models, and communicating complex findings to diverse stakeholders. I specialize in population analytics, spatial data science, and collaborative research projects with international organizations including USAID, WHO, and the University of Cambridge.

EDUCATION

Ph.D., Demography and Sociology

Expected May 2027

University of Pennsylvania, Philadelphia, PA

- Dissertation: Climate change impacts on maternal and perinatal health in Ghana.
- Affiliated student, International Max Planck Research School for Population, Health, and Data Science (IMPRS-PHDS).

M.A., Statistics and Data Science

Expected May 2026

University of Pennsylvania, Philadelphia, PA

- Coursework in advanced statistical modelling, machine learning, and applied data analysis.

M.Phil., Population and Health

2022

University of Cape Coast, Cape Coast, Ghana

B.A., Population and Health

2017

University of Cape Coast, Cape Coast, Ghana

ACADEMIC AND PROFESSIONAL APPOINTMENTS

Research Data Scientist

Aug 2023 – Present

USAID HEARTH Project, University of Pennsylvania

- Coordinate data integration and statistical analysis across multiple partner-country datasets, supporting research on linkages between health, environment, and agriculture.
- Develop and apply statistical and machine-learning models for health-outcome analysis and forecasting.
- Build interactive dashboards and visualizations using R Shiny and Power BI for monitoring and stakeholder communication.
- Mentor junior team members in survey methodology, statistical analysis, and reproducible workflow practices.

Health Educator (with data-analyst responsibilities)

Aug 2021 – Jul 2023

Ghana Health Service, Cape Coast Metropolitan Health Directorate, Ghana

- Designed and implemented health surveillance data systems serving district-level health facilities and a population of approximately 200,000.
- Developed predictive models for disease-outbreak detection and conducted statistical analysis of health-intervention effectiveness.

- Created data-visualization tools using Tableau and R for public health communication and district-level decision-making.
- Supported the Cape Coast Metropolitan Directorate in evidence generation for resource-allocation decisions across district health programs.

District Census Officer

Dec 2019 – Jul 2021

Ghana Statistical Service, 2021 Population and Housing Census

- Managed data collection and field implementation of the 2021 Population and Housing Census for a district of approximately 180,000 residents.
- Developed implementation plans aligned with national statistical frameworks and United Nations census guidelines.
- Supervised a team of more than fifty enumerators and field supervisors across more than 120 enumeration areas, ensuring standardized data-collection protocols and quality control.
- Built automated reporting systems using Python and Excel VBA for real-time progress monitoring during the enumeration period.

Graduate Teaching and Research Assistant

2017 – 2019

Department of Population and Health, University of Cape Coast, Ghana

- Supported undergraduate teaching in demography, population studies, and quantitative research methods.
- Contributed to faculty research projects on maternal health, vaccine uptake, and health-service utilization in Ghana.

PEER-REVIEWED PUBLICATIONS

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- Bediako, V. B.** (2025). Hidden by national averages: Unveiling the complex dynamics of ethnic disparities in under-five survival across sub-Saharan Africa. *SSM-Population Health*, 29, 101870.
- Bediako, V. B.**, et al. (2025). Factors associated with vaccine default in Southern Ghana based on data from the RTS, S malaria vaccine trial in Cape Coast. *Scientific Reports*, 15, 251.
- Okyere, J., **Bediako, V. B.**, et al. (2023). RTS, S/AS01E vaccine defaults in Ghana: A qualitative exploration of the perspectives of defaulters and frontline health-service providers. *Malaria Journal*, 22, 260.
- Ackah, J. A., Ayerakwah, P. A., Boakye, K., Owusu, B. A., **Bediako, V. B.**, Gyasi, M., et al. (2022). Circumcising daughters in Nigeria: To what extent does education influence mothers' FGM/C continuation attitudes? *PLOS Global Public Health*, 2(11), e0000660. <https://doi.org/10.1371/journal.pgph.0000660>
- Dickson, K. S., Okyere, J., Ahinkorah, B. O., Seidu, A.-A., Salihu, T., **Bediako, V.**, Owusu, B. A., Budu, E., Agbemavi, W., Edjah, J. O., & Darteh, E. K. M. (2022). Skilled antenatal care services utilisation in sub-Saharan Africa: A pooled analysis of demographic and health surveys from 32 countries. *BMC Pregnancy and Childbirth*, 22(1), 461. <https://doi.org/10.1186/s12884-022-05137-5>
- Bediako, V. B.**, Boateng, E. N. K., Owusu, B. A., & Dickson, K. S. (2021). Multilevel geospatial analysis of factors associated with unskilled birth attendance in Ghana. *PLoS ONE*, 16(6), e0253603. <https://doi.org/10.1371/journal.pone.0253603>
- Appiah, F., Owusu, B. A., Ackah, J. A., Ayerakwah, P. A., **Bediako, V. B.**, & Ameyaw, E. K. (2021). Individual and community-level factors associated with home birth: A mixed-effects regression analysis of the 2017–2018 Benin Demographic and Health Survey. *BMC Pregnancy and Childbirth*, 21, 547. <https://doi.org/10.1186/s12884-021-04014-x>
- Darteh, E. K. M., Dickson, K. S., Ahinkorah, B. O., Owusu, B. A., Okyere, J., Salihu, T., **Bediako, V. B.**, Budu, E., Agbemavi, W., Edjah, J. O., & Seidu, A.-A. (2021). Factors influencing the uptake of intermittent preventive treatment among pregnant women in sub-Saharan Africa: A multilevel analysis. *Archives of Public Health*, 79(1), 182. <https://doi.org/10.1186/s13690-021-00707-z>

Manuscripts in Preparation and Under Review

Several manuscripts on heat exposure and maternal–perinatal health in Ghana, conflict and child nutrition in Nigeria, and methodological replication of conflict-nutrition findings are in preparation or under review. Working titles available on request.

RESEARCH GRANTS AND CONSULTANCIES

Principal Investigator	2026 – 2027
<i>Penn Global Dissertation Grant, University of Pennsylvania</i>	
Health Systems, Climate Stress, and Early-Life Mortality in Ghana and Sub-Saharan Africa. Grant reference: PGDG-2026	
Co-Investigator	2025 – 2026
<i>Education Sub-Saharan Africa (ESSA) and REAL Centre, University of Cambridge</i>	
Project: Home-school language alignment, multilingualism, and reading skills among Primary 3 children in Ghana. Grant reference: FLAFRICA001.	
Principal Investigator	2022 – 2023
<i>World Health Organization and School of Public Health, University of Ghana</i>	
Project: Alternative implementation strategies for the administration of the fourth vaccine dose of the RTS, S in the Cape Coast Metropolis. Grant identifier: UGSPH/CIRT-M/001.	

AWARDS AND HONORS

Judith Rodin Fellowship	2026 – 2027
<i>Graduate Group in Sociology, University of Pennsylvania</i>	
PDRI–DevLab Graduate Student Research Grant	2026
<i>Penn Development Research Initiative – DevLab</i>	
Graduate Merit Award	2025
<i>Canadian Population Society Annual Meeting</i>	
Fishman Graduate Fellowship	2023 – 2024
<i>University of Pennsylvania</i>	
Benjamin Franklin Fellowship	2023 – 2024
<i>School of Arts and Sciences, University of Pennsylvania</i>	

INVITED TALKS AND CONFERENCE PRESENTATIONS

Selected presentations at the Population Association of America Annual Meeting, the Canadian Population Society Annual Meeting, the International Union for the Scientific Study of Population, and West Africa Political Economy (WGAPPE). A complete list is available on request.

TEACHING AND MENTORING

Mentor	2026 – Present
<i>Summer Undergraduate Mentored Research Program (SUMR), University of Pennsylvania</i>	
Mentored an undergraduate in statistical analysis, survey methodology, and data-visualization practices.	
Graduate Teaching Assistant	2017 – 2019
<i>Department of Population and Health, University of Cape Coast</i>	
Supported undergraduate teaching in demography, population studies, and research methods.	

SERVICE

Peer Review

Reviewer for BMJ Open, PLoS ONE, and Malaria Journal.

Professional Memberships

- International Union for the Scientific Study of Population (IUSSP)
- Population Association of America (PAA)
- Population-Environment Research Network (PERN)
- TDR Global – Special Program for Research and Training in Tropical Diseases

TECHNICAL SKILLS

Programming	R (advanced), Python (advanced), Stata (advanced), SQL (intermediate), LaTeX.
Statistical methods	Multilevel and hierarchical modelling; survival analysis; causal inference (target-trial emulation, instrumental variables); Bayesian methods (R-INLA, BYM2); machine learning; distributed lag non-linear models; time-series analysis.
Geospatial analysis	ArcGIS Pro, QGIS, GeoDa; spatial econometrics; geospatial linkage of survey data with environmental and conflict datasets.
Data visualization	Power BI, Tableau, R Shiny, ggplot2.
Survey and data tools	REDCap, ODK, SurveyCTO, CSPro.
Reproducible practice	Git, GitHub; literate programming with R Markdown and Quarto; parallel R workflows.

CERTIFICATIONS AND ADDITIONAL TRAINING

Social/Behavioral Research Certification <i>CITI Program</i>	2024
REDCap 101 OCR Certification	2024
Digital and Computational Demography <i>Max Planck Institute for Demographic Research</i>	2023
Spatial Data Science Certification <i>ESRI</i>	2022
Advanced GIS Analysis <i>University of California, Davis</i>	2021

REFERENCES

Available on request.